Cashew Disease Detection & XAI

Hybrid CNN-Transformer Integration

1. Methodology

This project utilizes a dual-branch hybrid architecture to detect cashew leaf diseases. The CNN branch (EfficientNet-B0) focuses on local texture and lesion extraction, while the Transformer branch (CaiT-XXS24) captures global spatial dependencies across the leaf surface.

2. Model Performance

The final Hybrid model achieved an outstanding accuracy of 99.02%. Comparative analysis showed this configuration outperformed individual CNN and Transformer baselines by an average margin of 2-3%.